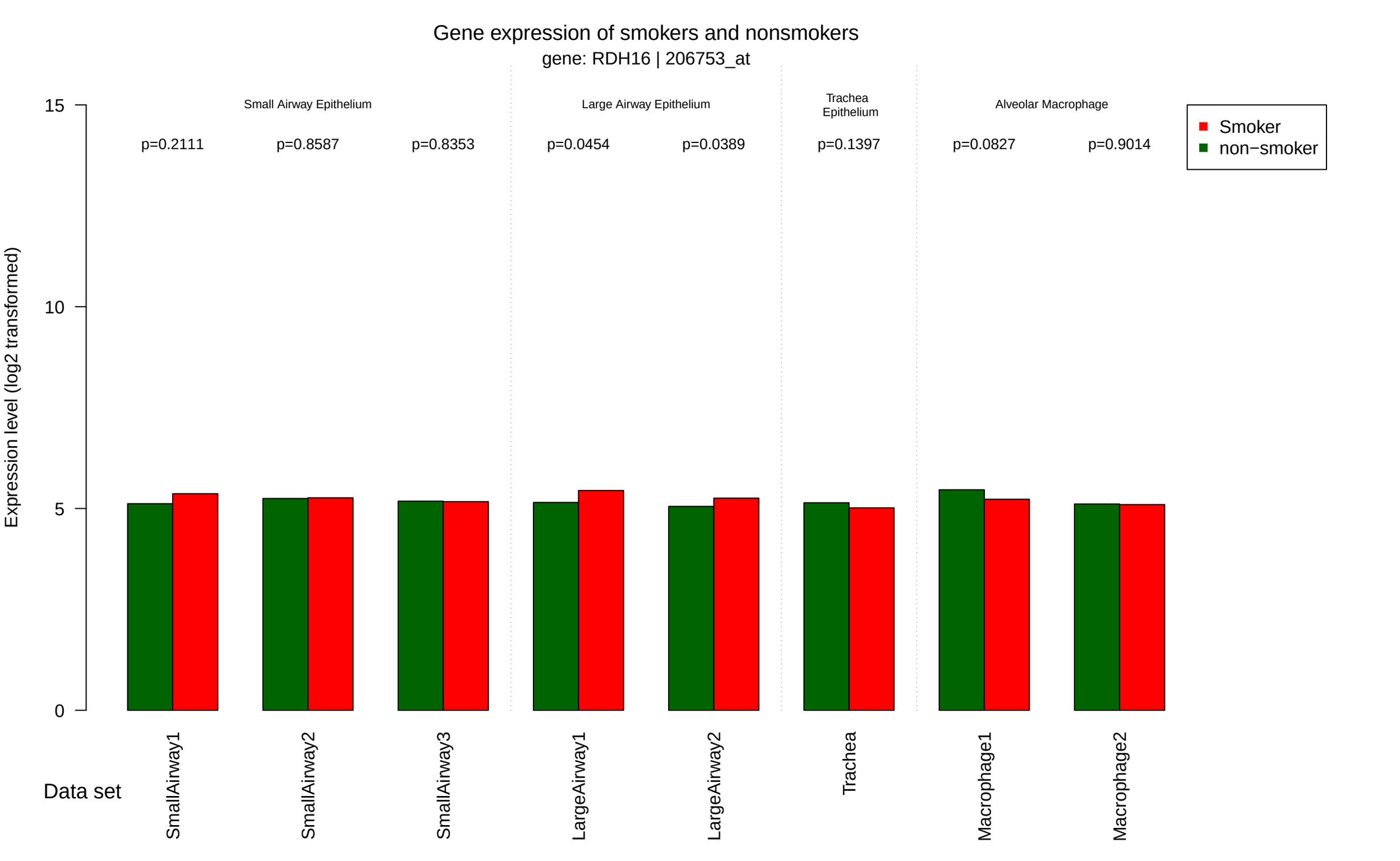
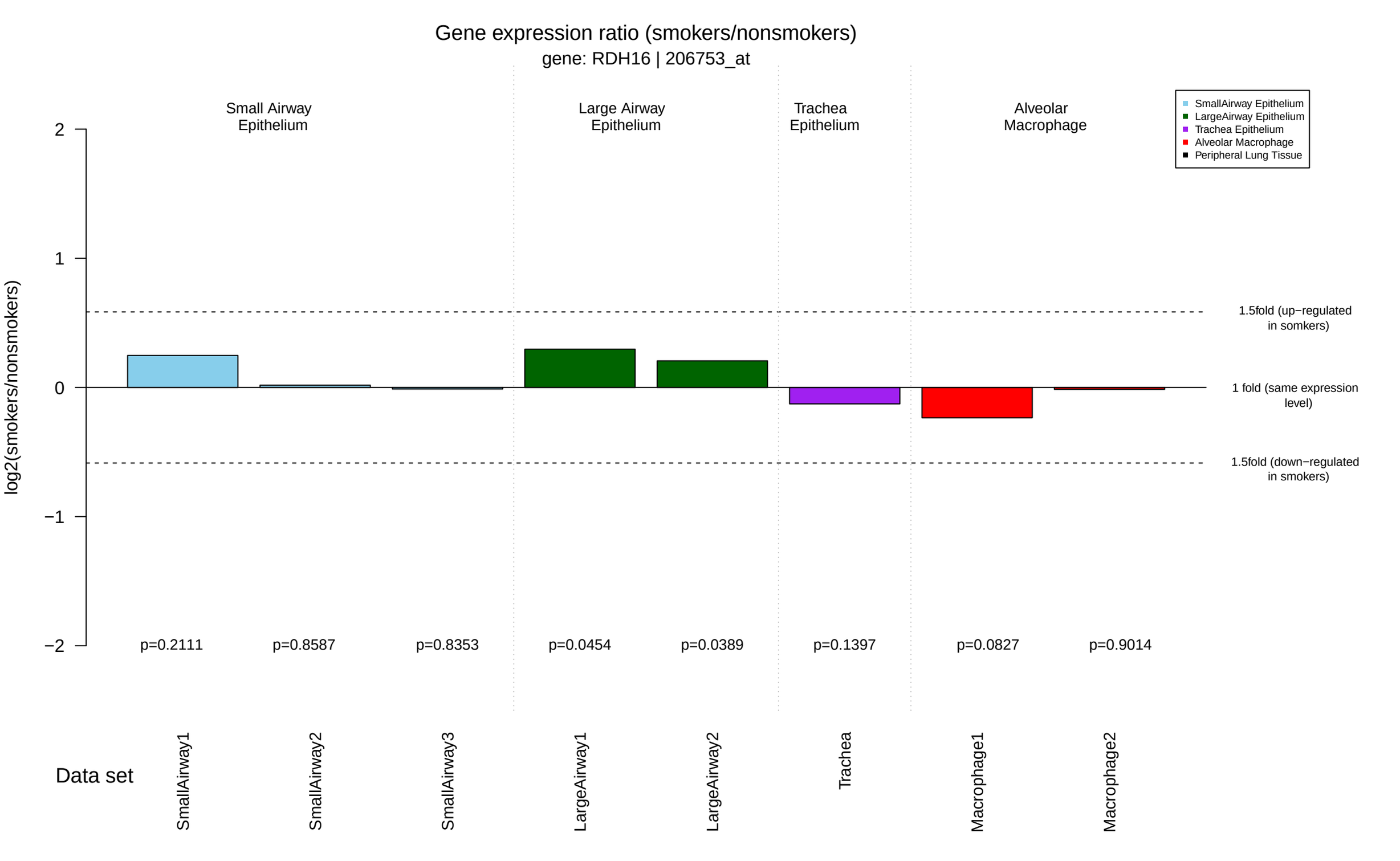
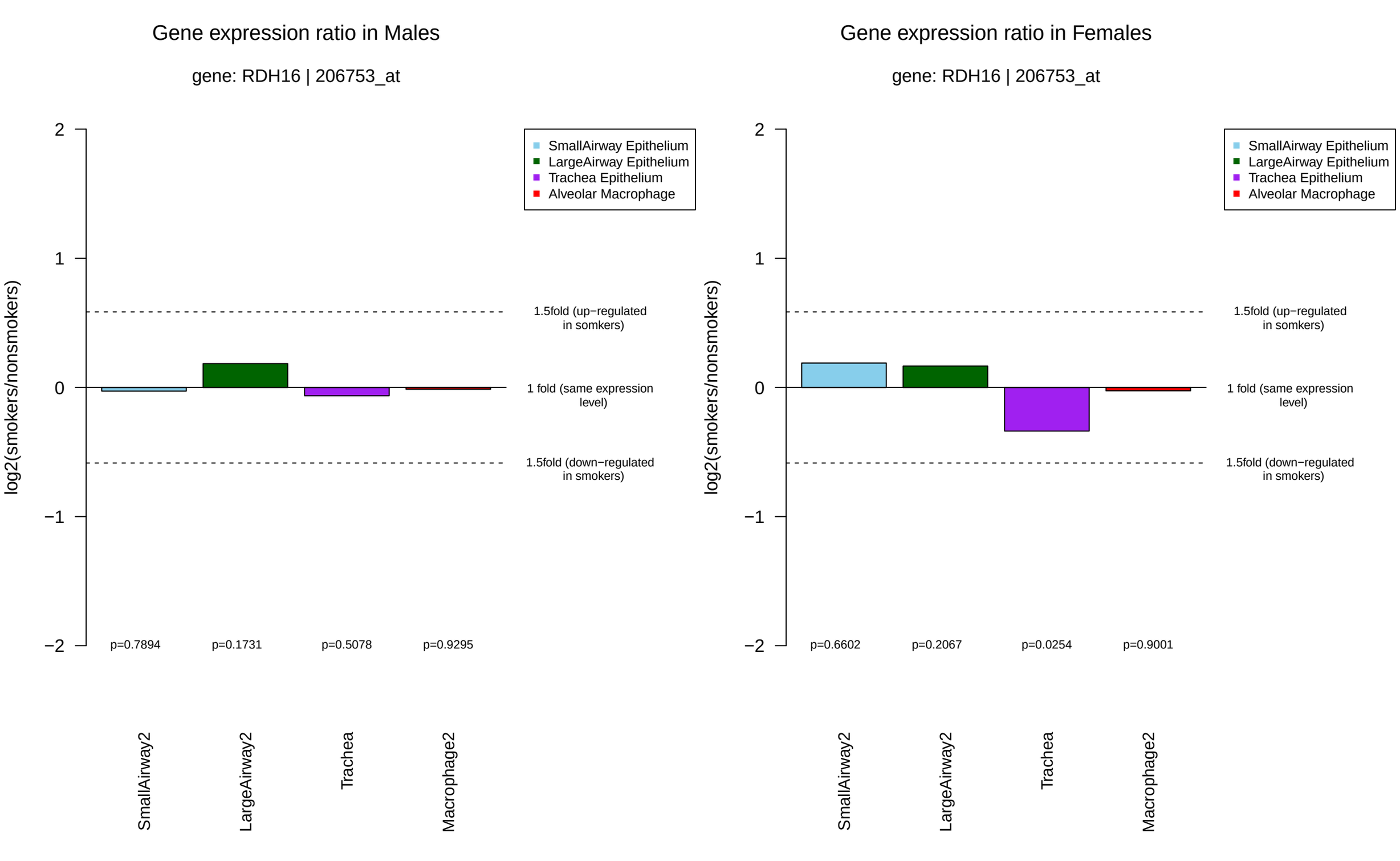
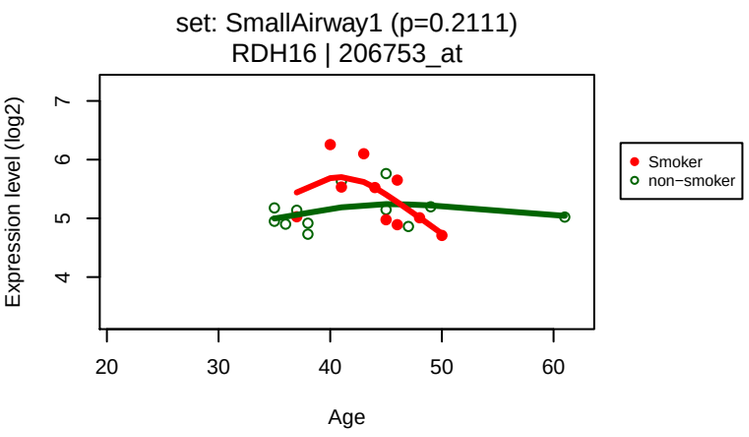
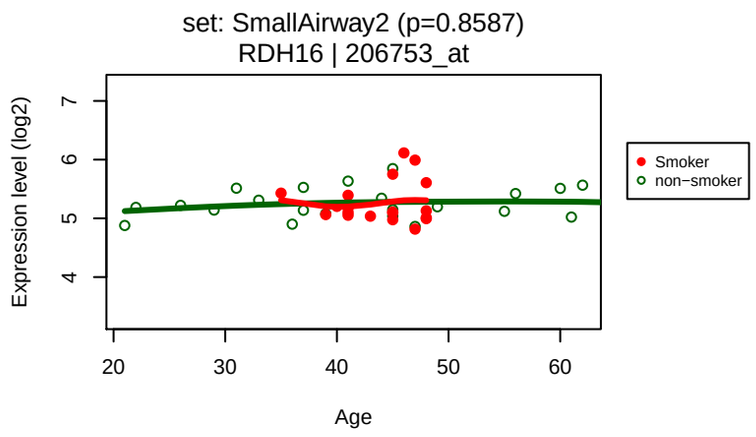
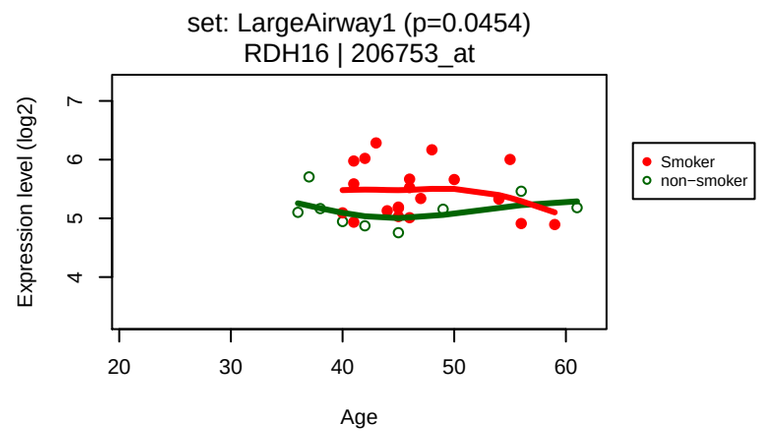

Gene expression of smokers and nonsmokers
gene: RDH16 | 206753_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2111
p=0.8587
p=0.8353
p=0.0454
p=0.0389
p=0.1397
p=0.0827
p=0.9014
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

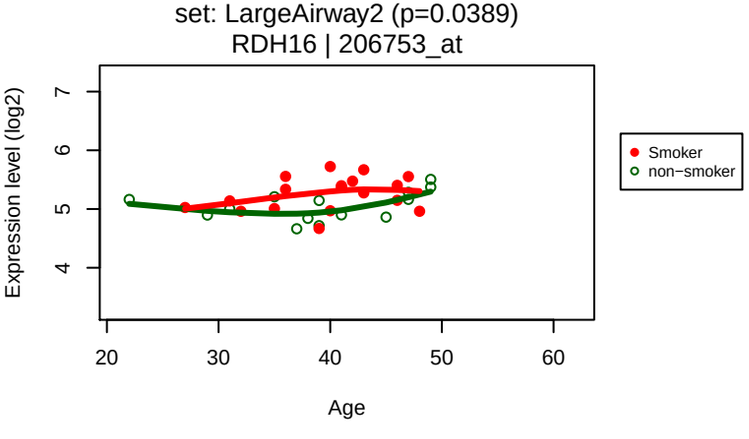

Gene expression ratio (smokers/nonsmokers)
gene: RDH16 | 206753_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2111
p=0.8587
p=0.8353
p=0.0454
p=0.0389
p=0.1397
p=0.0827
p=0.9014
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

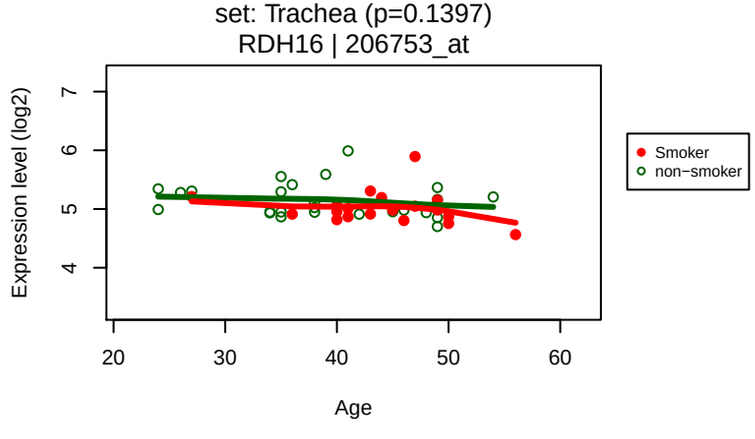

Gene expression ratio in Males
gene: RDH16 | 206753_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7894
p=0.1731
p=0.5078
p=0.9295
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RDH16 | 206753_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6602
p=0.2067
p=0.0254
p=0.9001
SmallAirway2
LargeAirway2
Trachea
Macrophage2

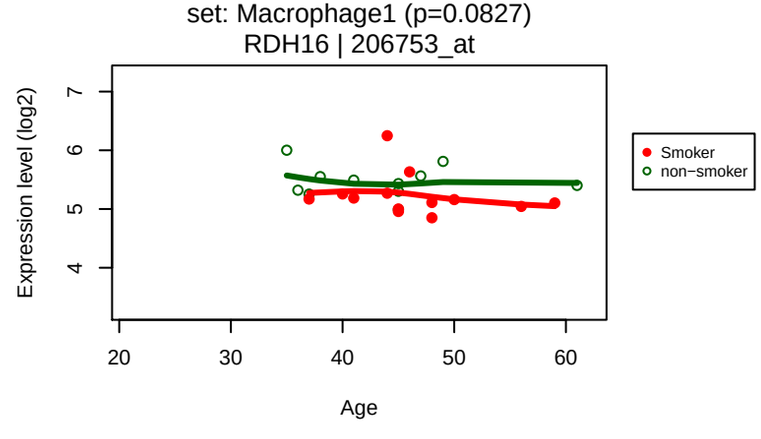

set: SmallAirway1 (p=0.2111)
RDH16 | 206753_at
Expression level (log2)
Age
Smoker
non−smoker

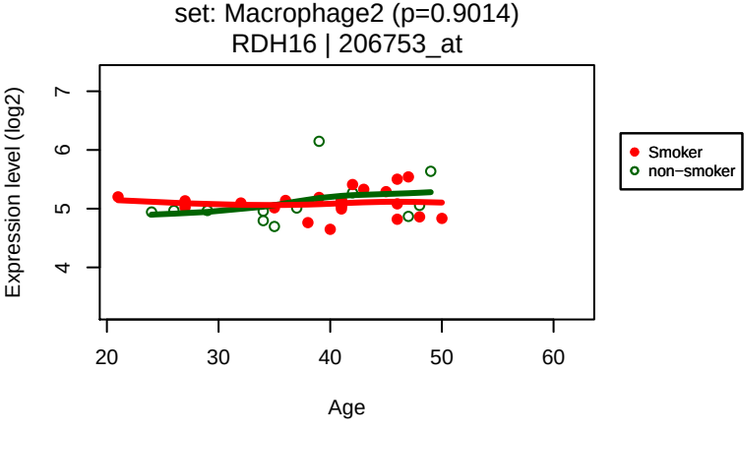

set: SmallAirway2 (p=0.8587)
RDH16 | 206753_at
Expression level (log2)
Age
Smoker
non−smoker

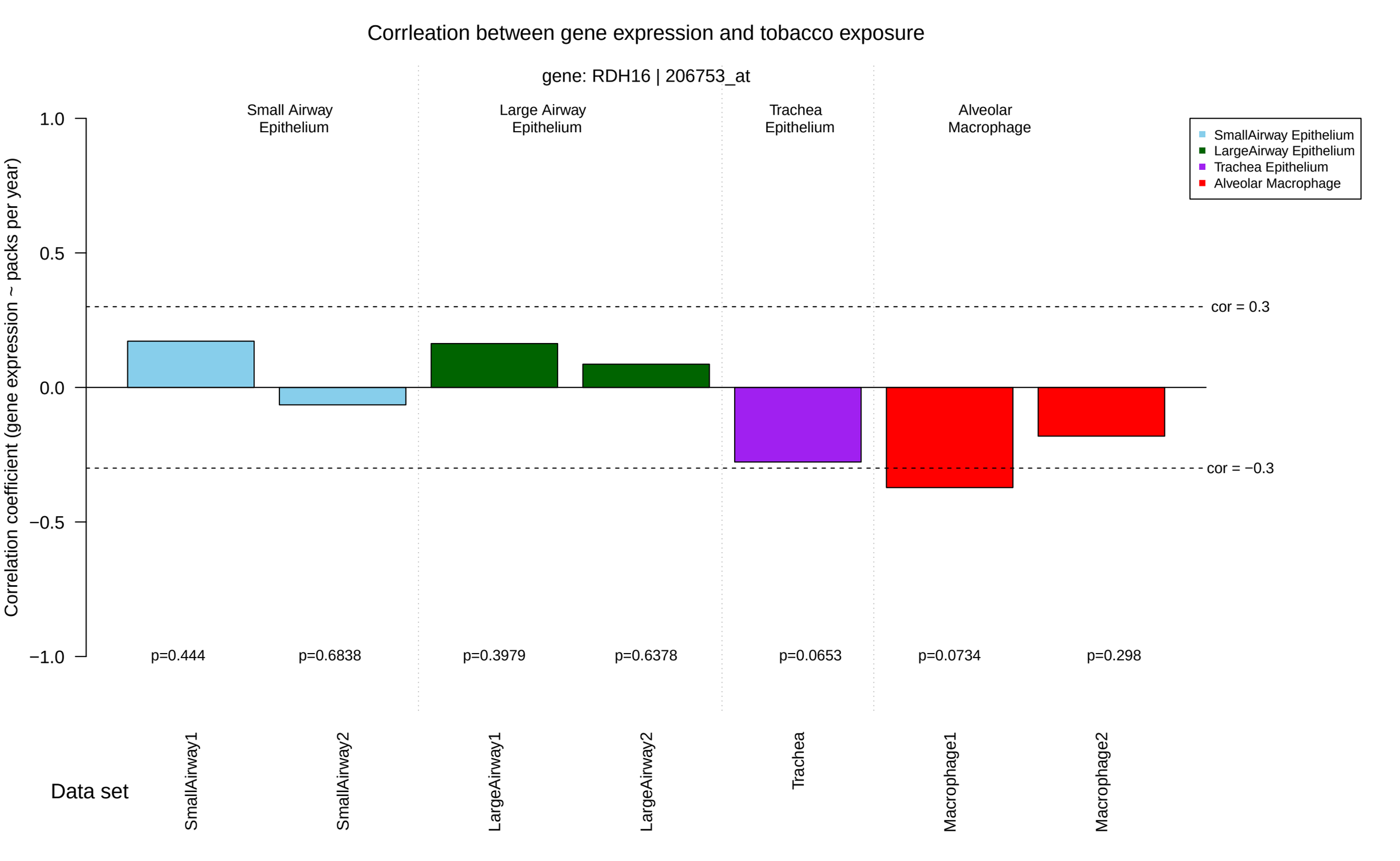

set: LargeAirway1 (p=0.0454)
RDH16 | 206753_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0389)
RDH16 | 206753_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1397)
RDH16 | 206753_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0827)
RDH16 | 206753_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9014)
RDH16 | 206753_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RDH16 | 206753_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.444
p=0.6838
p=0.3979
p=0.6378
p=0.0653
p=0.0734
p=0.298
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2